Eclipse Soundscapes Data Management: Pre-Eclipse Data Infrastructure & Deployment Preparation (Stage 0)

A methods and provenance reference for people using Eclipse Soundscapes (ES) datasets

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1. What This Document Covers

This document describes Stage 0 (Pre-Eclipse Infrastructure and Data Stewardship Planning) of the Eclipse Soundscapes (ES) data lifecycle. It documents the planning decisions, design principles, infrastructure systems, accessibility considerations, and operational workflows that were developed before the October 14, 2023 annular solar eclipse and the April 8, 2024 total solar eclipse. These decisions established the foundation for all later stages of the ES data stewardship pipeline.

ES is a NASA-funded participatory science project that deployed AudioMoth acoustic recorders during two eclipses: the October 14, 2023 annular solar eclipse and the April 8, 2024 total solar eclipse. Volunteers recorded environmental soundscapes across the United States so that researchers could study how rapid eclipse-related light changes affected animal behavior and ambient sound.

Stage 0 focused on designing the systems and workflows needed to support large-scale, geographically distributed participatory science data collection. Unlike the later Stage documents, which focus on how submitted data moved through processing, validation, public archiving, and scientific analysis workflows, this document explains why the underlying infrastructure and operational decisions were made and how those decisions shaped Stages 1 through 4.

This document is intended for researchers, archivists, developers, coordinators, and participatory science practitioners interested in understanding how ES approached scalable, accessibility-focused, privacy-aware, and open-science-oriented infrastructure design. The goal is to document the operational realities, design tradeoffs, and implementation lessons learned so that future projects can build upon these experiences rather than recreating them independently.

2. What Stage 0 Produced

Stage 0 produced the operational infrastructure, metadata systems, deployment workflows, and participant preparation materials that enabled the ES data lifecycle to function across the 2023 and 2024 eclipse campaigns.

- **Prepared and Distributed Data Collector Kits**
  Standardized AudioMoth deployment kits prepared for participant use, including configured devices, microSD cards, batteries, printed instructions, accessibility modifications, and return materials.
- **Master Data Collectors Spreadsheet**
  Centralized spreadsheet linking ES ID #s with AudioMoth serial numbers and supporting downstream metadata reconciliation, validation, processing, and public archiving workflows.
- **Metadata Collection System**
  Online and written metadata collection workflows used to gather recording locations, timing information, and participant-provided site notes associated with each ES ID #.
- **Participant Training and Deployment Materials**
  Data Collector training manuals, deployment instructions, accessibility guidance, and timing procedures supporting standardized field data collection.

- **Pre-Eclipse Data Stewardship and Processing Infrastructure**
  Initial planning and workflow development supporting downstream intake, processing, validation, public archiving, and scientific analysis workflows.

These deliverables established the operational and organizational foundation for all later stages of the ES data lifecycle.

3. Why Stage 0 Was Needed

Stage 0 was necessary because ES relied on thousands of distributed participants collecting scientifically usable audio data across a geographically dispersed eclipse path. Supporting this type of large-scale participatory science effort required substantial infrastructure development before data collection could begin.

The project needed systems for:

- assigning and tracking AudioMoth devices,
- linking devices to site-level identifiers,
- collecting metadata consistently,
- preparing and distributing recording kits,
- supporting accessible participation,
- and planning downstream workflows for data intake, validation, processing, archiving, and analysis.

Because ES was designed to support both scientific usability and broad public participation, infrastructure development also needed to account for varying levels of technical experience, internet access, accessibility needs, and deployment environments.

Stage 0 established the operational foundation required for all later stages of the Eclipse Soundscapes (ES) data lifecycle. Unlike later stages, which primarily focused on organizing, processing, validating, or sharing participant-submitted data, Stage 0 focused on the pre-eclipse decisions, systems, supports, and preparation tasks that made data collection possible. The workflow diagram below highlights the major Stage 0 activities that directly or indirectly shaped ES data collection, including equipment and protocol decisions, ES ID infrastructure, metadata submission systems, accessibility supports, participant training materials, device and kit preparation, and kit distribution. The complete ES Operational Infrastructure and Data Stewardship Workflows diagram is included in Appendix 1.

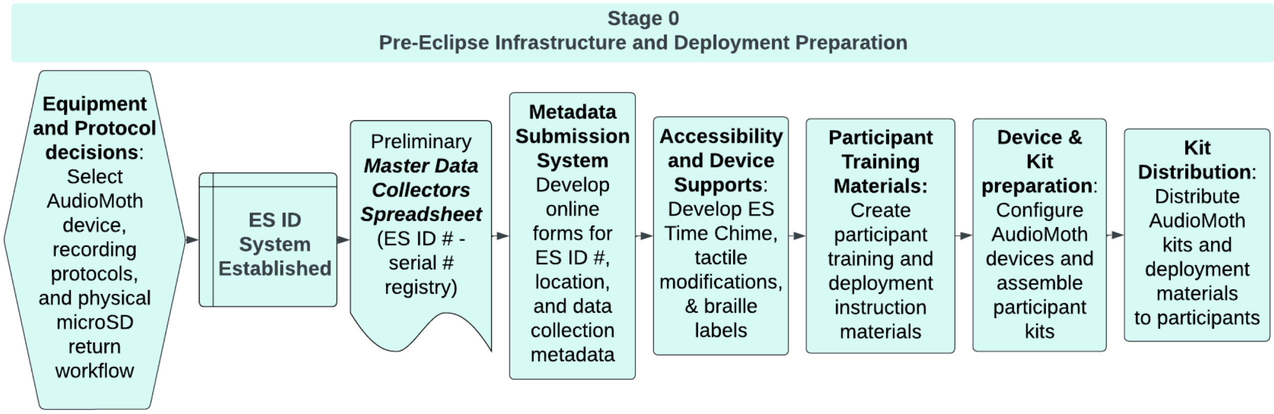

Figure 1*. Eclipse Soundscapes Stage 0 workflow for pre-eclipse infrastructure development and deployment preparation. Unlike later stages that focused on transforming and processing participant datasets, Stage 0 established the operational systems, metadata infrastructure, accessibility supports, participant training materials, and deployment workflows required to support large-scale participatory science data collection during the 2023 and 2024 eclipse campaigns.*

4. The 2023 Annular Eclipse as a Beta Test

It is a rare and fortunate occurrence to have two solar eclipses occur on the same continent within a short time frame. Two eclipses occurring in the United States within a year provided an opportunity for the ES team to test and then improve its processes. The audio data collection procedures designed for the October 14, 2023 annular solar eclipse served as the beta test for the April 8, 2024 total solar eclipse. However, the 2023 experiment was also a genuine scientific undertaking: recording devices were deployed, data were collected, and results contributed to the project’s scientific goals. It was also the first real-world stress test of the logistical systems that the team had built.

The smaller geographic footprint and lower participation levels associated with the 2023 annular eclipse made it a more manageable context in which to identify infrastructure limitations and workflow failures before the larger 2024 campaign. While early operational planning projected approximately 200–300 Data Collector participants across both eclipse campaigns, the project was intentionally designed with the understanding that participation could exceed these estimates. As a result, many infrastructure and workflow decisions were evaluated not only for immediate feasibility, but also for their ability to scale to substantially larger levels of public participation if interest exceeded expectations. Across the 2023 and 2024 eclipse campaigns, ES ultimately supported nearly 1,000 registered Data Collector participants and generated approximately 25 TB of raw audio data. Hundreds of these recording sites later progressed through the full multi-stage validation, processing, and public archival workflow described throughout the remaining Stage documents.

As a beta test, it was expected that changes would be made to the 2024 program based on lessons learned during the 2023 implementation. Any significant changes to ES procedures or experiment protocols are highlighted throughout this document.

5. Audio Data and its Collection Requirements

Planning began with an understanding of both the scientific requirements for the audio data and the volunteer scientist's needs for participating as Data Collectors.

Scientific Audio Data Needs:

- High-quality audio data for each recording site
- Latitude and longitude for each recording site
- Time-stamped audio data
- NASA eclipse timing data integration

Volunteer Scientist Needs:

- Free or inexpensive equipment
- Easy-to-use equipment and clear data collection protocols
- No or low bandwidth requirements for data sharing
- Both visually and tactilely accessible equipment
- Website(s) and application(s) that meet Web Content Accessibility Guidelines (WCAG)
- Clear instructions and training
    - Not explained in this report. This can be found in:
      Severino, M., Winter, H., & Bauer, D. J. (2026). Eclipse Soundscapes Data Collector Role Training and Implementation Manual (2023–2024). Zenodo. https://doi.org/10.5281/zenodo.18623443
- Ability to find or track individual submissions publicly without using personally identifiable information (PII)

Together, these requirements established the foundational design constraints that shaped all Stage 0 infrastructure decisions. The project needed to support scientifically useful, time- and location-specific audio collection while remaining accessible, low-cost, low-bandwidth, privacy-aware, and scalable for a geographically distributed novice volunteer community. These combined requirements directly influenced equipment selection, device deployment workflows, metadata collection strategies, accessibility modifications, the design of the Eclipse Soundscapes Identification system, and long-term data stewardship planning described throughout the remainder of this document. One of the earliest and most consequential infrastructure decisions involved determining how large volumes of participant-collected audio data could be returned, stored, and shared without creating participation barriers for volunteers.

6. Data Submission Constraints and Implementation Decision

Because each AudioMoth deployment generated approximately 40–60 GB of audio data, the method of participant data submission became a major infrastructure, accessibility, and long-term data stewardship consideration during early project planning. The project required a workflow that could support geographically distributed participation without imposing high-bandwidth internet requirements, complex technical upload procedures, or barriers to long-term public access and reuse of the resulting datasets.

6.1 Cloud-based Upload Option and Cost Analysis

The initial Eclipse Soundscapes proposal envisioned using Arbimon as the primary long-term platform for acoustic data storage, browsing, and analysis. At the time of proposal development, Arbimon provided free storage for large acoustic datasets along with integrated acoustic analysis tools, creating the possibility that participants and researchers could both access and analyze recordings directly within the same platform.

As project planning progressed following award notification, the ES team began evaluating how participant-generated audio data could realistically be transferred into such a system at national scale. Early planning initially assumed that volunteers would upload recordings directly to Arbimon. However, operational testing and workflow evaluation revealed several challenges. Upload permissions to shared datasets required manual approval workflows, creating substantial administrative overhead for a geographically distributed volunteer community.

To address these limitations, the ES team began exploring a hybrid architecture in which participants would upload recordings to Amazon Web Services (AWS) infrastructure, after which the ES team would download, curate, and transfer datasets into Arbimon for long-term storage and analysis access. AWS services including Elastic Compute Cloud (EC2) and S3 Glacier Deep Archive were evaluated to support upload handling, temporary storage, processing workflows, and raw data management.

At the same time, the ES team worked with project partners and community-serving organizations to evaluate whether large-scale participant upload workflows would be practical in real-world deployment settings. Library partners in particular expressed concerns that high-bandwidth upload requirements would create substantial barriers both for library staff and for many community participants, especially in rural or bandwidth-limited areas.

These concerns aligned with the ES team's own testing of upload feasibility under typical household internet conditions.

6.2 Volunteer Self-Upload Testing and Infrastructure Limitations

In parallel with infrastructure planning, the ES team evaluated the practical feasibility of participant-led uploads using projected dataset sizes and average household upload speeds.

Testing demonstrated that upload times for a single AudioMoth dataset (~60 GB) would create substantial barriers for many participants. At an average U.S. household upload speed of approximately 75 Mbps, a full upload would require nearly two uninterrupted hours. Slower or unstable connections increased upload times significantly and raised the likelihood of failed or interrupted transfers.

Upload speed	Estimated time for 60 GB
100 Mbps	~1 hr 20 min
75 Mbps (Average US Household Speed 2020)	~1 hr 47 min
50 Mbps (5G good)	~2 hr 40 min
25 Mbps (5G fair)	~5 hr 20 min
15 Mbps (5G)	~9 hr 29 min

6 Mbps (5G low)	~22 hr 14 min
5 Mbps (Average US Household Speed 2008)	~26 hr 40 min

Table 1. Estimated upload times for a 60 GB AudioMoth dataset at various connection speeds. The average 2008 US household upload speed of 5 Mbps was used as a planning baseline.

These limitations disproportionately affected participants with limited, rural, mobile-only, library-based, or unreliable internet access and conflicted directly with the project's goal of broad, geographically distributed participation.

At the same time, the long-term sustainability of the proposed cloud architecture became increasingly uncertain. During this planning period, Arbimon transitioned organizationally to Rainforest Connection. As the platform evolved, previously open or freely available APIs and workflows began to change, making it increasingly difficult to determine how large Eclipse Soundscapes datasets could be programmatically transferred into the platform and whether participants and researchers would continue to have unrestricted analysis access. It also became unclear whether raw audio files could be downloaded freely at large scale, which represented a major concern for the project's open-science goals.

Because of these uncertainties, the ES team also evaluated whether AWS alone could serve as the complete upload, storage, and distribution infrastructure for the project. At the time of planning, projected participation levels suggested a total raw audio dataset size of approximately 8.6–12.9 TB. Analysis using the AWS Pricing Calculator indicated that baseline archival storage alone was not the primary barrier for a dataset of this scale. However, AWS download and data distribution costs introduced a much larger uncertainty. Because the project's goals required that datasets remain freely accessible and downloadable to the public, long-term retrieval and large-scale public access costs became difficult to predict and potentially unsustainable.

The analysis ultimately highlighted a broader data stewardship challenge: the project required not only a place to store large datasets, but also a sustainable long-term strategy for upload handling, public access, unrestricted download, reuse, and open-science preservation. Implementing and maintaining custom cloud upload and processing infrastructure would also have introduced substantial long-term technical complexity and operational overhead.

6.3 Data Submission Decision: Physical Media Return

Based on the combined infrastructure, accessibility, bandwidth, and long-term data stewardship challenges identified during project planning, the ES team determined that large-scale participant self-upload workflows were not appropriate for Eclipse Soundscapes data collection.

Instead, the project adopted a physical media return model using mailed microSD cards. Participants receiving ES Data Collector Kits were provided with pre-addressed return envelopes, while participants using their own AudioMoth devices were instructed to mail their microSD cards directly to ARISA Lab using return information provided through the Eclipse Soundscapes website.

This approach eliminated high-bandwidth upload requirements and substantially reduced technical barriers for participants, supporting broader and more geographically distributed participation. It also ensured that the ES team retained a physical copy of the raw data while continuing to evaluate sustainable long-term archival and open-access data-sharing solutions.

The physical return workflow became a foundational component of the broader Eclipse Soundscapes data stewardship pipeline and directly influenced kit design, metadata collection procedures, physical return logistics, and downstream processing workflows described throughout the remainder of this document.

Following additional evaluation of long-term archival and open-access options, including exploration of NASA repository pathways, the project ultimately adopted Zenodo as the long-term public archive for Eclipse Soundscapes datasets. The reasoning behind this later transition and the development of the Zenodo publication workflow are described in the Stage 3 documentation:

- Winter, H., & Severino, M. (2026). Eclipse Soundscapes Data Management: Public Audio Data Sharing (Stage 3). Zenodo. https://doi.org/10.5281/zenodo.18683437

7. Equipment Selection: The AudioMoth Acoustic Data Recorder

With the physical data return workflow established, the project next required a recording device that could meet four non-negotiable criteria simultaneously:

1. Inexpensive enough to distribute at scale to volunteer scientists across the United States
2. Capable of producing science-grade acoustic recordings
3. Simple enough for a non-specialist volunteer to configure, deploy, and return
4. Open-source and well-documented to support training, troubleshooting, and long-term archiving

The AudioMoth (Open Acoustic Devices, openacousticdevices.info), described in the scientific literature by Hill et al. (2019), was selected as the primary data collection device for both the 2023 and 2024 eclipse campaigns because it was one of the few available recorders that met all four criteria without modification.

7.1 Cost and Scale

At the time of project planning, commercially available passive acoustic monitoring (PAM) systems commonly used in ecological research, including the SongMeter series from Wildlife Acoustics and the BAR series from Frontier Labs, were typically priced at approximately $800 USD per unit or higher (Hill et al., 2019). At that price point, deploying hundreds of devices across the continental United States to cover the full range of eclipse intensities required by the project's science questions would have been cost-prohibitive.

In contrast, the AudioMoth is a credit-card-sized printed circuit board (58 × 48 × 15 mm) built around an ARM Cortex M4 microcontroller and a micro-electro-mechanical systems (MEMS) microphone. At the time of project planning, a single assembled AudioMoth unit typically cost approximately $80–125 USD, depending on order quantity, making it roughly an order of magnitude less expensive than entry-level commercial PAM alternatives (Hill et al., 2019). This cost profile made it feasible to procure and distribute hundreds of units to volunteer data collectors at no cost to the volunteer, with enough budget remaining to support training, packaging, and logistics.

7.2 Scientific Validity

The AudioMoth was not chosen solely for its cost. By the time of the ES:CSP grant award, the device had already been used and validated in published scientific studies across a range of acoustic monitoring applications, including soundscape analysis, bat echolocation detection, and bird survey work (Hill et al., 2019; Darras et al., 2019). The device records uncompressed audio to a microSD card at sample rates from 8,000 to 384,000 samples per second, spanning audible wildlife sounds through bat-range ultrasound, and embeds a UTC timestamp directly into each WAV filename and file metadata (AudioMoth Operation Manual, 2022). These timestamps, together with site-specific location data, enabled recordings to be correlated with eclipse contact times during analysis, making accurate temporal and geographic metadata among the most critical components of the entire dataset.

Standard acoustic indices used in soundscape ecology, the scientific framework that was the initial, unused basis for ES's approach to measuring changes in animal behavior, can be calculated directly from uncompressed WAV recordings at the sample rates the AudioMoth produces. The device's full-spectrum capability also meant that a single hardware configuration could capture sound across the frequency ranges relevant to birds, insects, amphibians, and bats, without requiring species-specific hardware customization for each deployment site.

The recording and metadata decisions made during Stage 0 later supported scientific analysis of how animals responded to the 2023 and 2024 eclipses. Because each recording included accurate timestamps, location information, and site-level identifiers, researchers were able to connect audio recordings to eclipse timing data and other ecological datasets during later analysis work. Subsequent stages of the ES data lifecycle preserved, validated, organized, and publicly archived these datasets in ways that supported downstream scientific interpretation and reproducibility.

One example is a study examining changes in bird vocal behavior during the eclipses:

- Gilbert, N. A., Pease, B. S., Severino, M., & Winter III, H. (2026). Photic niche explains avian behavioral responses to solar eclipses. Ecology and Evolution, 16(2), e73090. https://doi.org/10.1002/ece3.73090

The supporting geographic data, eclipse timing information, processed analysis outputs, figures, and open analysis code associated with this work were also publicly archived through the companion Zenodo record:

- Gilbert, N., & Pease, B. (2026). Eclipse Soundscapes Analysis Data and Reproducible Materials for Bird Vocalization Responses to the 2023 and 2024 Solar Eclipses (eclipse-traits). Zenodo. https://doi.org/10.5281/zenodo.19112632

7.3 Ease of Use (Usability)

Ease of use was a critical requirement given the project's reliance on volunteer scientists with a wide range of technical backgrounds. The AudioMoth's simple physical interface, including a three-position switch controlling power and recording modes, and its ability to operate for extended periods without continuous user interaction made it well-suited for distributed data collection.

Once activated, the device could record continuously throughout the deployment period without requiring ongoing monitoring or complex interaction from participants. This simplicity reduced the likelihood of deployment errors and allowed volunteers to focus primarily on deployment procedures, metadata collection, and retrieval of the device after the observation period. The relatively simple physical workflow also reduced the amount of technical training and supporting equipment required for participation compared to many traditional passive acoustic monitoring systems.

More detailed discussion of configuration constraints, clock-setting workflows, and accessibility considerations is provided in Section 6.

7.4 Open-Source Design

The AudioMoth hardware, firmware, and supporting applications are all released under a Creative Commons Attribution 4.0 International license. Complete design files, source code, and documentation are publicly available through the Open Acoustic Devices GitHub repository (github.com/OpenAcousticDevices). This openness was directly aligned with the ES's long-term data management goals, enabling complete documentation of recording specifications and configuration parameters in project data dictionaries and archival records without reliance on proprietary vendor systems.

Summary of Selection Rationale

Unit cost	~\$80–\$125 USD (vs. ~\$800+ for entry-level commercial PAM devices)
Scientific validity	Validated in peer-reviewed studies prior to ES:CSP deployment
Open source	CC BY 4.0; hardware, firmware, and software documentation fully public
Sample rates	8–384 kHz; full-spectrum coverage from birds through bat ultrasound
Recording format	Uncompressed WAV with UTC timestamp in filename and file metadata

Although the AudioMoth met the project's primary scientific, financial, and usability requirements, its use in a large-scale volunteer deployment introduced several technical and operational constraints that required additional workflow design and device modification decisions. These constraints, and the solutions developed to address them, are described in the following section.

8. Device Constraints and Implementation Decisions

8.1 Device Configuration Complexity (Default vs Custom)

The AudioMoth offers three operating modes — USB/OFF, CUSTOM, and DEFAULT — controlled by a physical three-position switch (AudioMoth Operation Manual, 2022). From a purely scientific perspective, the “CUSTOM” mode would have been preferable because it would have allowed recordings to be scheduled around eclipse periods rather than producing large volumes of continuous audio data outside the primary analysis window.

However, use of the CUSTOM mode required interaction with the AudioMoth Configuration App[1], which was not WCAG compliant at the time of the project, highly technical, and not user friendly for first-time users. Unlike the clock-setting workflow, for which the ES team was able to develop a partially accessible workaround through the Eclipse Soundscapes Time Chime, there was no practical way for the project to simplify or make the broader device configuration process meaningfully accessible within the scope of the project.

In addition, CUSTOM mode settings depended on the device retaining power after configuration. The project’s initial deployment model required participants to install batteries themselves prior to deployment, and battery removal or power loss could reset the AudioMoth’s internal clock and erase configuration-dependent settings. Because clock synchronization was already recognized as an unavoidable technical friction point within the deployment workflow, the ES team intentionally avoided introducing an additional participant-facing configuration process that could require repeated troubleshooting, retraining, or repeated technical setup by volunteers.

As a result, while the CUSTOM mode better aligned with scientific recording efficiency, it would have introduced substantial accessibility barriers, technical complexity, and ongoing participant support requirements. Based on these constraints, the DEFAULT recording mode was selected as the standard configuration because it better supported reliable volunteer participation while still meeting the project’s scientific requirements.

8.1.1 Simplified Default Settings

The DEFAULT setting records at a sample rate of 48 kHz with gain set to Medium. Under the Nyquist sampling principle, a recording system can accurately capture frequencies up to approximately half of its sampling rate. At a 48 kHz sampling rate, this means sounds at approximately 24 kHz and below can be reliably recorded. Because the average range of human hearing is approximately 20 Hz to 20 kHz, this configuration allowed the AudioMoth to detect and record many bird, insect, and other environmental vocalizations within and slightly beyond the range of human hearing.

During early project planning, the ES team anticipated that volunteer scientists might participate directly in listening to recordings and helping identify audible animal sounds within the datasets. For this reason, ensuring that recordings captured frequencies within the human hearing range was considered important during protocol development. By the time large-scale analysis work began advances in machine learning-based acoustic analysis workflows made large-scale volunteer listening and manual sound identification less necessary than initially anticipated.

On the DEFAULT settings, the AudioMoth begins recording immediately upon the switch being set to the DEFAULT position. Once started on DEFAULT, the AudioMoth produces a series of

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[1] https://www.openacousticdevices.info/applications

continuous WAV-format recording files containing approximately 12 hours of audio each. This meant that volunteers could switch the AudioMoth to DEFAULT at deployment and leave it unattended until the end of the recording protocol.

Because the DEFAULT setting eliminated the need for participant-led configuration, it better supported reliable novice volunteer participation. However, continuous recording introduced additional constraints related to data volume, power usage, and storage, which are addressed in the following sections.

8.2 Time Setting Constraints and Solutions

The AudioMoth's internal clock assigns a UTC timestamp to each recorded audio file. Whenever batteries are removed, the device resets its internal real-time clock to 01/01/1970 at 00:00 (AudioMoth Operation Manual, 2022). As a result, the clock must be reset whenever batteries are installed or replaced. Because audio data analysis depended on aligning recordings with site-specific eclipse timing and participant location metadata, accurate timestamp generation was one of the most critical technical requirements of the entire data collection workflow.

At the time of protocol development, two primary methods existed for setting the AudioMoth clock:

- connecting the device to a computer and using the AudioMoth Configuration Application
- using the Open Acoustic Devices "time chime" website (https://chime.openacousticdevices.info/) to play an audio signal that synchronizes the device clock

The AudioMoth Configuration Application required connecting the device to a computer via USB, was not WCAG compliant during ES project execution, and presented a complex interface unsuitable for many novice users. The alternative acoustic time chime method reduced technical complexity by allowing the device to synchronize its clock through an audio signal played from a webpage. However, at the time of ES project planning, the Open Acoustic Devices time chime website was also not WCAG compliant.

An additional device-specific limitation was that the AudioMoth confirmed successful synchronization only through flashing LED lights, creating accessibility barriers for participants who are blind or have low vision and causing confusion for many sighted participants as well.

These limitations required the ES team to develop alternative clock-setting workflows that balanced accessibility, usability, and scientific timing requirements.

8.2.1 ES Time Chime Site and Accessibility Limitations

To address accessibility limitations in the existing AudioMoth clock-setting workflow, the ES team developed a WCAG-compliant webpage called the Eclipse Soundscapes Time Chime. The page played an audio signal that the AudioMoth could detect through its microphone, allowing volunteers to synchronize the device clock without requiring direct interaction with the AudioMoth Configuration Application. The ES Time Chime page was designed to meet WCAG 2.1 Level AA compliance standards, and accompanying training materials provided accessible step-by-step instructions for participants.

However, while the ES Time Chime improved accessibility of the clock-setting process itself, the AudioMoth device itself still confirmed successful clock setting using flashing LED lights. Participants who are blind or have low vision could not independently verify successful synchronization using this visual confirmation system, and many sighted participants also found the flashing light patterns difficult to interpret reliably.

To partially mitigate this issue during the 2023 deployment, the ES Time Chime audio was configured to play twice automatically because device clock setting did not always succeed on the first attempt.

These concerns directly influenced additional metadata collection procedures. Participants were encouraged during the 2023 deployment to document and share the date and time at which recording began so that the ES team could later compare participant-reported start times against AudioMoth timestamps during data validation. After the 2023 beta deployment demonstrated the value of these participant-provided notes for recovering or interpreting recordings affected by clock-related issues, written date and time documentation became a required component of the 2024 protocol.

More information on training related to the Time Chime is available:

- Severino, M., Winter, H., & Bauer, D. J. (2026). Eclipse Soundscapes Data Collector Role Training and Implementation Manual (2023–2024). Zenodo. https://doi.org/10.5281/zenodo.18623443
- Severino, M. (2026, February 11). Eclipse Soundscapes Training: How to set the time chime on your AudioMoth device (Data Collector Role). Zenodo. https://doi.org/10.5281/zenodo.18614955

8.3 Data Volume and Storage Planning

Using the default sampling rate of 48kHz with continuous recording and medium gain, an AudioMoth produces two 4295 MB files per day, for a total of 8590 MB (~8.59 GB) per day of recording. To capture both eclipse-period recordings and non-eclipse baseline conditions before and after the event, the project adopted a five-day recording window consisting of two recording days on either side of the eclipse. This resulted in an estimated recording size of ~43 GB per deployment. A 64 GB SanDisk Extreme UHS Speed Class 3 (U3) microSDXC card was selected as sufficient for the experiment while providing additional margin if the AudioMoth was deployed early. These estimates were initially calculated using AudioMoth configuration tools and then verified by the ES team through experimental testing under real-world conditions.

During project planning, ES anticipated approximately 200–300 Data Collector participants across the 2023 and 2024 eclipse campaigns. Based on the projected recording size per deployment, the expected total data volume was estimated at approximately ~8.6–12.9 TB of raw audio data. These calculations informed microSD card selection, storage planning, and broader infrastructure considerations needed to support large-scale continuous acoustic recording across distributed observation sites. These constraints directly influenced battery and deployment planning, discussed in Section 6.4.

Actual participation and recording volume ultimately exceeded these early planning estimates, contributing to a final archive size of approximately 25 TB of raw audio data across both eclipse campaigns.

8.4 Power Requirements and Battery Performance

8.4.1 Power Usage

With the AudioMoth configured for continuous recording over a minimum five-day data collection period, battery life was a critical consideration. A loss of power at any point during the data collection window would have rendered the dataset from that site unusable.

At the DEFAULT recording settings, the AudioMoth consumes approximately 280 milliamp-hours (mAh) per day, or 1400 mAh over a five-day period (AudioMoth Operation Manual, 2022). The device is powered by three AA batteries connected in series, providing 4.5 volts direct current (VDC), which falls within the AudioMoth's operating range of 3.5–6 VDC. Recording ceases when voltage drops below approximately 3.4 VDC.

Because the batteries are connected in series, the total capacity remains equivalent to a single battery (~2700 mAh), resulting in an estimated margin of approximately 700 mAh, or an additional ~2.7 days of recording beyond the required observation period. These estimates were generated using the AudioMoth configuration app and verified through experimental testing, in which devices deployed under real-world conditions continued recording for more than 10 days before falling below operational voltage thresholds.

These results confirmed that standard AA battery configurations provided sufficient power for the full observation period with an additional safety margin.

8.4.2 Deployment Strategy and Iteration (2023–2024)

Despite confirming that AA batteries could support the full recording period, an additional operational decision was required regarding how batteries would be distributed and installed.

The 2023 deployment strategy was intentionally designed to support participation in both the 2023 annular eclipse and the 2024 total solar eclipse using the same AudioMoth device. Most kits therefore included enough materials for both eclipse campaigns, including two microSD cards, two plastic deployment bags, six AA batteries, four zip ties, and two return mailers.

For the 2023 deployment, AudioMoth devices were shipped with batteries uninstalled because the ES team was concerned that inactive devices might slowly consume battery power prior to deployment. However, this decision also required participants to install batteries and complete device clock setting themselves shortly before recording began.

As discussed in Section 6.2.1, participant-led clock setting became one of the most common sources of confusion and support requests during the 2023 deployment. In addition to accessibility limitations associated with interpreting the AudioMoth LED confirmation patterns, many participants were uncertain whether the device clock had been set successfully.

Following the 2023 deployment, additional testing demonstrated that inactive AudioMoth battery usage while powered off was negligible and did not significantly affect recording duration.

Based on these findings, the 2024 deployment workflow was modified so that project-distributed devices were shipped with batteries pre-installed and clocks synchronized in advance by the ES team. This substantially reduced reliance on participant-led clock setting and improved overall deployment consistency.

Participants only needed to repeat the clock-setting process if batteries became dislodged or removed prior to deployment. As a result, additional troubleshooting materials and training resources related to clock setting remained part of the 2024 participant support workflow.

8.5 Lack of Built-in GPS and Location Strategy

The Eclipse Soundscapes project used AudioMoth version 1.2.0 devices, which were the most current AudioMoth model available at the time of procurement. Devices were ordered early because of global supply chain concerns, the need for sufficient testing time, and the importance of standardizing hardware across deployments so that participant training could be developed around a consistent device model. AudioMoth version 1.2.0 did not include built-in Global Positioning System (GPS) capability for automatically recording location data.

The absence of integrated GPS functionality presented a significant limitation because accurate latitude and longitude coordinates were critical for correlating recordings with site-specific eclipse timing and coverage conditions. Automatic location capture would have reduced participant burden and improved metadata consistency across deployments.

During project planning, the ES team evaluated the AudioMoth GPS Board, an expansion board designed to add GPS functionality to AudioMoth devices and synchronize recordings using GPS time. While compatible with AudioMoth version 1.2.0 devices, implementation would have required additional GPS hardware costs and manual soldered installation on each device. Because hundreds of AudioMoth devices were being prepared for deployment, this approach would have introduced substantial additional labor, technical complexity, and additional points of failure into an already highly manual preparation workflow.

At the same time, project planning and procurement occurred during significant global supply chain disruptions in 2021. The ES team needed to secure sufficient hardware early in the project timeline to allow time for testing, protocol development, and creation of standardized participant training materials using a consistent device model. Waiting for potential future AudioMoth versions with integrated GPS capability was therefore not considered operationally feasible for the project timeline.

Based on cost, complexity, hardware availability, and deployment scalability considerations, participant-provided GPS coordinates were adopted as the standard location strategy for the project.

8.6 Weatherproofing Constraints and Implementation Decision

The AudioMoth is an exposed circuit board attached to a 3 × AA battery holder and is not inherently weatherproof. As a result, a protective enclosure was required for outdoor deployment to prevent water damage during the recording period.

Two primary options were considered:

- The official AudioMoth IPX7 Waterproof Case (Open Acoustic Devices)
- A low-cost alternative using sealed polypropylene bags

The official IPX7 case is an injection-molded polycarbonate enclosure with a compression O-ring seal and a Porelle acoustic vent that allows sound transmission while maintaining waterproofing (AudioMoth Operation Manual, 2022). While this solution provided high-quality environmental protection, its cost at the time of project planning (~$45 USD per unit) would have substantially increased the per-kit cost, limiting the number of AudioMoth devices the ES project could purchase and provide to volunteer scientists. Although the design files are open-source and available for 3D printing, producing and assembling hundreds of cases was determined to be both costly and time-intensive.

As an alternative, the use of water-tight polypropylene bags was evaluated. These bags are commercially available, commonly used with AudioMoth devices, and recommended by the device developers (Hill et al., 2017). Prior studies indicate that while such enclosures may reduce the recording of ultrasonic frequencies, frequencies within the range relevant to the Eclipse Soundscapes investigation are largely unaffected (Darras et al., 2019).

To validate this approach, the ES team conducted comparative field testing by deploying AudioMoth devices simultaneously in both polypropylene bags and official IPX7 cases under identical conditions. Representative audio samples were analyzed using Raven Lite software (Cornell Lab of Ornithology), including both auditory review and spectrogram analysis. No discernible differences were observed in the frequency ranges relevant to the project.

Based on cost, scalability, and testing results, polypropylene bags were selected as the primary weatherproofing solution for device deployment.

8.7 Accessibility Modifications

The AudioMoth's low-cost design uses a minimal physical interface consisting primarily of a single three-position slide switch and small silk-screened labels printed directly on the circuit board. The use of simple tactile physical controls, rather than touchscreen or app-dependent interaction, supported ease of use for many participants and reduced the complexity of basic device operation. However, the small size of the switch and printed labels still created usability challenges, particularly for participants who are blind or have low vision. As a result, additional tactile labeling and wayfinding modifications were developed by the ES team to improve device orientation and operation for a wide range of users.

8.7.1 Tactile Modifications

To support consistent device orientation and non-visual wayfinding, tactile markers were added to key locations on each AudioMoth device distributed by the ES team.

Silicone bump dots of specific sizes and shapes were applied in a standardized pattern, with each size and shape consistently assigned to a particular switch, port, or device feature. This standardized tactile mapping allowed participants to identify controls and orient the device by touch in a repeatable and learnable way. While these modifications were particularly important for participants who are blind or have low vision, the tactile cues also supported device wayfinding and usability for participants more broadly because of the AudioMoth's small physical controls and printed labels.

Clear bump dots were selected to ensure that printed labels beneath them remained visible, preventing the tactile modifications from obscuring visual text and supporting participants with varying levels of vision.

The tactile cues were incorporated directly into the standard Data Collector instructions and training materials used by all participants rather than being presented as a separate accessibility resource. This helped establish a consistent shared orientation system across the project and reduced reliance on visual confirmation during deployment and operation.

More information on training related to tactile device modifications is available:

- Severino, M., Winter, H., & Bauer, D. J. (2026). Eclipse Soundscapes Data Collector Role Training and Implementation Manual (2023–2024). Zenodo. https://doi.org/10.5281/zenodo.18623443

8.7.2 Braille Labeling

Braille labeling was implemented to support participants who are blind to independently identify their assigned Eclipse Soundscapes Identification Number (ES ID #), the project's site-level tracking identifier discussed further in Section 7, on both the AudioMoth device and its packaging. This was a critical component of enabling independent participation in the data collection process.

Prior to the 2023 deployment, multiple approaches to creating braille labels were evaluated, including the use of puff paint and a handheld braille labeler. Puff paint was initially considered due to its low cost and availability; however, testing revealed several limitations, including inconsistent dot spacing, difficulty producing accurate braille patterns, and uncertainty about long-term durability after drying.

Based on these limitations, a handheld embossed braille labeler was selected for the 2023 deployment. This method provided more consistent dot spacing and improved legibility compared to puff paint. However, issues with the labeling approach did not become apparent immediately. Over time, the vinyl label tape retained curvature from storage, and its adhesive backing proved insufficient for reliable attachment to the AudioMoth device and packaging. After several days, labels began to detach during handling.

To address this issue during the 2023 deployment, cyanoacrylate adhesive (CA glue) was applied to secure each label after placement. While this improved adhesion, it introduced additional challenges, including the risk of glue interfering with braille readability if applied improperly.

These limitations informed modifications to the labeling approach for the 2024 deployment. The project transitioned to the use of a 6dot braille labeling system with more flexible, adhesive-backed label tape. This system improved ease of production, label durability, and overall usability, reducing the need for additional adhesive and improving reliability during shipping and field deployment. This iterative approach to braille labeling reflects the broader project strategy of refining accessibility solutions based on real-world deployment feedback.

9. ES ID System: Device Registration and Backend Implementation

Stage 0 established the foundational identification and tracking system used throughout the Eclipse Soundscapes (ES) data lifecycle. This system centered on the assignment of a unique Eclipse Soundscapes Identification Number (ES ID #) to each data collection device and the creation of the Preliminary *Master Data Collectors Spreadsheet*, which served as the central system of record for all subsequent stages.

At its core, the ES ID system provided a persistent site-level identifier used to associate AudioMoth devices, returned recordings, and participant-submitted metadata throughout the Eclipse Soundscapes workflow, while maintaining separation between participant identity and publicly shared datasets.

9.1 Preliminary *Master Data Collectors Spreadsheet*

The Preliminary *Master Data Collectors Spreadsheet* was created during Stage 0 as the authoritative registry linking ES ID #s to AudioMoth device serial numbers.

At this stage, the spreadsheet functioned as:

- A device registration log

- A site-level identifier system
- The foundation for all downstream metadata integration

No site-level metadata (e.g., location, timing, or participant notes) were incorporated at this stage. Those data were collected and integrated during Stages 1 and 2.

- Severino, M., & Winter, H. (2026). Data Management: Receipt, Sorting, and Metadata Organization (Stage 1). Zenodo. https://doi.org/10.5281/zenodo.19471425
- Winter, H., & Severino, M. (2026). Eclipse Soundscapes Data Management: Data Processing (Stage 2). Zenodo. https://doi.org/10.5281/zenodo.18683402

9.2 ES ID System Role Across the Data Lifecycle

The ES ID system established in Stage 0 served as the site-level linkage framework connecting downstream processing, metadata validation, audio analysis, and public data publication workflows across all subsequent stages.

Specifically:

- In **Stage 1**, the *Preliminary Master Data Collectors Spreadsheet* was expanded to incorporate participant-submitted location data, derived site-specific eclipse timing data, and sorting status, forming the central metadata record for each site. Participant-submitted metadata collected through online forms used ES ID #s to associate location and site information with each recording device. When online metadata submissions were missing or incomplete, handwritten notes returned with the microSD cards were manually reviewed and used to recover or validate participant-provided location information.

  Once location information was confirmed, the ES-developed *Eclipse Phase Timing Tool (EPTT)* and related programmatic workflows were used to match each recording site with NASA Goddard Space Flight Center eclipse prediction data based on geographic coordinates. These derived eclipse timing and eclipse coverage values were then added to the spreadsheet alongside the participant metadata and ES ID # site record.

  - Severino, M., & Winter, H. (2026). Data Management: Receipt, Sorting, and Metadata Organization (Stage 1). Zenodo. https://doi.org/10.5281/zenodo.19471425
- In **Stage 2**, the ES-developed *ES WAV Audio Validation & Extraction Suite (ES WAVES)* was used to ingest and process returned audio data, verify AudioMoth serial numbers embedded in CONFIG.TXT files and WAV metadata, and confirm the linkage between audio recordings and ES ID # site records. The ES-developed *ES Audio Metadata Evaluation System (ES AMES)* was then used to evaluate embedded recording timestamps and identify recordings requiring additional timestamp review or validation. Handwritten participant notes documenting recording start dates and times, submitted either online or on paper forms, were manually reviewed when needed to help validate or interpret recordings from devices with clock-related issues. Participant-reported timing information was then manually incorporated into the Master Data Collectors Spreadsheet where available.
  - Winter, H., & Severino, M. (2026). Eclipse Soundscapes Data Management: Data Processing (Stage 2). Zenodo. https://doi.org/10.5281/zenodo.18683402

- In **Stage 3**, each ES ID # served as the unit of dataset organization, with one site-level dataset and corresponding Zenodo record created per ES ID #. The ES-developed *Automated Zenodo Upload Software (AZUS)* was used to package site-level data and metadata, generate standardized Zenodo documentation, and automate upload workflows through the Zenodo API.
    - Winter, H., & Severino, M. (2026). Eclipse Soundscapes Data Management: Public Audio Data Sharing (Stage 3). Zenodo. https://doi.org/10.5281/zenodo.18683437

Through this design, **the three-digit ES ID # functioned as the persistent unifying identifier across physical materials, digital audio files, metadata records, analysis workflows, and published datasets.**

9.3 Privacy and Data Separation

The ES ID system was designed to support privacy-aware data management while maintaining traceability across the data lifecycle.

The *Master Data Collectors Spreadsheet* served as the central internal record and included both participant-related information (used for project operations, communication, and follow-up) and site-level metadata associated with each ES ID #. The ES ID system enabled downstream processing, analysis, and public data sharing workflows to operate using site-level identifiers rather than participant identity.

In practice:

- ES ID #s were used as the primary reference for data handling, processing, analysis, and dataset organization workflows
- Participant identity information was not required for downstream audio processing, metadata validation, or analysis workflows
- Publicly shared datasets were organized, identified, and searchable by ES ID #, while personally identifiable information (PII) remained separate from publicly released metadata.

This approach ensured that datasets remained traceable to a specific recording site while protecting participant privacy in all publicly released data products.

Further details on how metadata and site-level records were handled during receipt, processing, validation, and public publication are provided in the Stage 1, Stage 2, and Stage 3 documentation:

- Severino, M., & Winter, H. (2026). Data Management: Receipt, Sorting, and Metadata Organization (Stage 1). Zenodo. https://doi.org/10.5281/zenodo.19471425
- Winter, H., & Severino, M. (2026). Eclipse Soundscapes Data Management: Data Processing (Stage 2). Zenodo. https://doi.org/10.5281/zenodo.18683402
- Winter, H., & Severino, M. (2026). Eclipse Soundscapes Data Management: Public Audio Data Sharing (Stage 3). Zenodo. https://doi.org/10.5281/zenodo.18683437

9.4 ES ID # Assignment and Device Registration

Each AudioMoth device used in the project, whether distributed by the ES team or registered by participants using their own devices, was assigned a unique three digit ES ID # before deployment.

During this process:

- The AudioMoth device serial number was recorded
- A corresponding three digit ES ID # was assigned
- The ES ID #–serial number pairing was entered into the Preliminary Master Data Collectors Spreadsheet

This registration process established the foundational ES ID #–serial number relationships used throughout subsequent processing and validation workflows.

9.4.1 Project-distributed Device Registration Workflow

For project-distributed devices, this manual assignment process occurred before shipment and required manual registration of each AudioMoth device. Each unit was connected to a laptop via USB, its serial number was accessed using the AudioMoth Config application, and that value was recorded in the Preliminary *Master Data Collectors Spreadsheet* alongside the next available ES ID #. This process was repeated for each device, resulting in a complete registry linking all deployed AudioMoth units to their assigned ES ID numbers.

Because this work was performed manually and at scale, it represented a time-intensive but critical step that enabled all downstream data linkage, validation, and processing.

9.4.2 Participant-Owned Device Registration Workflow

For the 2024 total solar eclipse, participants were given the option to register and use their own AudioMoth devices. To support this, ES ID assignment for participant-owned devices was managed through a project-developed online registration system designed to extend the ES ID framework beyond project-distributed equipment.

To maintain a clear distinction between internally distributed devices and participant-owned devices, ES ID numbers assigned through the online system began at 600. ES ID numbers below this range were reserved for project-distributed AudioMoth units that were manually registered before shipment.

Through the online interface, participants entered their AudioMoth device serial number, which triggered the assignment of a unique ES ID #. The backend system recorded the serial number, participant email address, and assigned ES ID # in a structured dataset. This dataset was then manually integrated into the *Master Data Collectors Spreadsheet*, ensuring that all devices, regardless of origin, were incorporated into the same centralized metadata framework.

The registration system was designed to enforce uniqueness, preventing duplicate ES ID assignments once a number had been issued. Upon registration, participants were immediately shown their ES ID # on-screen and received a confirmation email containing the same information. This dual-delivery approach ensured that participants could reliably reference their ES ID # during metadata submission and data return, even if the number was not recorded initially. This addition for 2024 data collection expanded participation pathways while maintaining consistency in downstream data processing and validation workflows.

10. 2023 Pre-Deployment AudioMoth Preparation and Training

Although Eclipse Soundscapes reduced some participant-facing technical complexity through the use of standardized DEFAULT mode settings and pre-configured devices, participation as a Data Collector still required substantial training related to deployment procedures, clock synchronization, waterproofing methods, metadata documentation, and physical data return workflows.

Participant support included written guides, training videos, live webinars, Q&A sessions, and step-by-step instructional materials designed for volunteers with a wide range of technical backgrounds.

Detailed participant training information is documented separately in:

- Severino, M., Winter, H., & Bauer, D. J. (2026). Eclipse Soundscapes Data Collector Role Training and Implementation Manual (2023–2024). Zenodo. https://doi.org/10.5281/zenodo.18623443

In parallel with participant training, the ES team implemented a standardized internal preparation workflow prior to device distribution.

While all project-distributed devices were AudioMoth version 1.2.0, firmware versions varied at the time of acquisition. To ensure consistency across testing, training, and deployment workflows, all project-distributed AudioMoth devices were flashed to firmware version 1.8.1 prior to deployment. Participants using their own AudioMoth devices were instructed through training materials to use the same firmware version.

The following preparation steps were performed for each project-distributed AudioMoth device:

- A 64GB SanDisk Extreme UHS Speed Class 3 (U3) microSDXC card was inserted into the device
- The AudioMoth was connected to a computer via USB and placed into USB mode
- Using the AudioMoth Flash App, the device was flashed to firmware version 1.8.1
- Using the AudioMoth Config App, the device serial number was verified and associated with the assigned ES ID # in the Preliminary Master Data Collectors Spreadsheet (see Section 6)
- The assigned ES ID # was written in large, legible text directly on the AudioMoth device and its corresponding box to support visibility and traceability
- Following technical preparation and ES ID assignment, devices were transferred into the kit assembly workflow described in Section 9.

This standardized preparation workflow ensured consistency across project-distributed devices and supported reliable metadata linkage, accessibility modification workflows, and downstream data processing.

11. 2023 Data Collector Kit Preparation and Distribution

Following device preparation (Section 8), Eclipse Soundscapes (ES) Data Collector Kits were assembled to provide participants with all materials required for device deployment, metadata submission, and microSD card return. The ES ID # assigned during device setup served as the central identifier linking the AudioMoth device, return materials, metadata records, and downstream data processing workflows.

Kits were assembled manually using a standardized workflow designed to support consistency across all distributed devices and materials. Each kit contained:

- 1 AudioMoth Device
- 1 waterproof plastic bag
- 3 zip ties (to attach AudioMoth to tree/pole),
- 1 padded envelope with ES ID # and return label affixed (to mail ES Team the MicroSD card for processing),
- 3 AA batteries taped together.
- one 64GB Extreme UHS Speed Class 3 (U3) microSDXC card
- 1 business card containing a URL and QR code (printed and embossed in braille) linking to Eclipse Soundscapes training materials and step-by-step instructions on the ES website

This kit structure was designed to support low-cost, accessible, large-scale deployment while minimizing participant technical burden.

11.1 Instruction Delivery Design Decision

Rather than including printed instruction manuals in each kit, the ES team chose to provide training and deployment instructions through online resources linked from a business card included in every package.

Providing large-print and braille instruction packets for each kit was considered; however, this approach would have substantially increased printing, assembly, and mailing costs while also limiting the ability to revise instructions in response to participant feedback or evolving procedures.

The business card approach reduced material costs, simplified kit assembly, and allowed participants to access the most current instructions and training materials on the Eclipse Soundscapes website. The cards included printed text, braille labeling, and a QR code to support multiple access methods.

This approach aligned with the project’s emphasis on accessibility, scalability, and iterative improvement.

11.2 Accessibility Modifications and Labeling

Before packaging, each project-distributed AudioMoth device received participant-facing accessibility modifications designed to improve usability for participants with a range of visual abilities.

These modifications included:

- Application of silicone bump dots in standardized tactile locations
- Braille labels displaying the assigned ES ID # on both the device and its packaging

These accessibility modifications were applied prior to packaging as part of the standardized kit preparation workflow described in Section 6.7.

11.3 Kit Assembly Workflow

Kit assembly followed a structured manual workflow in which all materials were organized in advance and assembled sequentially. Mailing envelopes, return envelopes, waterproof bags, batteries, zip ties, business cards, and labeling materials were prepared in batches prior to final kit assembly.

The workflow included:

- Preparing and labeling return envelopes
- Preparing participant mailing labels
- Adding standardized kit components to each mailing envelope
- Adding the corresponding AudioMoth device and microSD card
- Verifying kit completeness prior to shipment

Because each kit required manual preparation, labeling, and verification, the process was labor-intensive even at the 2023 deployment scale.

11.4 Kit Assignment and Distribution Workflow

Data Collector Kits were distributed through two primary pathways: partner-based distribution and direct individual assignment.

11.4.1 Partner-Based Distribution

Partner organizations, including groups such as the Nationwide Eclipse Ballooning Project, NASA at My Library, and subject matter expert (SME) collaborators, received kits in grouped ES ID # ranges. Kits were packaged together and shipped in bulk to a designated partner contact for local distribution.

The ES team co-developed and delivered training sessions and webinars tailored to partner communities; however, all participants used the same core training materials and procedures.

11.4.2 Individual Distribution

Individual participants applied to receive free Data Collector Kits from the Eclipse Soundscapes team. For accepted participants, mailing information was associated with the assigned ES ID # in the Master Data Collectors Spreadsheet, and mailing labels were generated from this record system.

This workflow ensured that each distributed device remained associated with a consistent site-level identifier throughout all later stages of data processing and archiving.

11.5 Packaging and Distribution

Once assembled and assigned, kits were packaged and shipped through the United States Postal Service (USPS). Individual kits were mailed directly to participants, while partner distributions were consolidated into bulk shipments where appropriate.

At the 2023 deployment scale, shipment preparation and in-person mailing at local post offices represented a significant operational effort. Individual package handling, transportation, and postal processing required substantial staff time and highlighted scalability limitations that informed later workflow changes implemented for the 2024 eclipse campaign (see Section 10).

12. Addressing Data Collection Protocol Issues post-2023 Beta Testing

The 2023 annular eclipse served as the first full-scale operational test of the Eclipse Soundscapes data collection system. This deployment revealed several operational, accessibility, and metadata-management challenges that were not fully apparent during protocol development and small-scale testing. These findings informed a series of targeted updates implemented for the 2024 total solar eclipse to improve data reliability, accessibility, scalability, and participant support workflows.

12.1 Participant Support and Instructional Delivery Improvements

During the weeks leading up to the 2023 eclipse deployment, the ES team experienced a substantial increase in participant questions related to deployment procedures, recording workflows, and protocol timing. Many of these questions were already addressed within the project's training videos, written instructions, and website documentation, indicating that participants often benefited from receiving guidance in smaller, time-relevant segments aligned with the deployment timeline.

In response, participant support and instructional reinforcement were expanded for the 2024 deployment. Updates included additional live Q&A sessions, expanded office hours in the weeks leading up to the eclipse, and the introduction of a Mailchimp-based drip email campaign. These scheduled emails provided volunteers with weekly reminders, deployment tips, and step-by-step guidance corresponding to what participants should be preparing or completing at that stage of the deployment process. The emails reinforced existing training materials while presenting information in smaller, time-relevant segments.

More information on training workflows and instructional materials is available in:

- Severino, M., Winter, H., & Bauer, D. J. (2026). Eclipse Soundscapes Data Collector Role Training and Implementation Manual (2023–2024). Zenodo. https://doi.org/10.5281/zenodo.18623443

12.2 AudioMoth Clock Setting and Timestamp Reliability

As discussed in Section 6, participant-led device clock setting remained one of the most significant sources of deployment difficulty during the 2023 campaign. Stage 2 processing later demonstrated that missing or incorrect timestamps affected the usability of some recordings for scientific analysis.

To improve timestamp reliability for the 2024 deployment, several protocol changes were implemented:

- participants were required to record and submit their recording start date and time both through the online form and as a handwritten note included with the returned microSD card
- project-distributed AudioMoth devices were shipped with batteries pre-installed and clocks synchronized in advance by the ES team
- additional troubleshooting guidance and training resources related to clock setting were incorporated into participant support materials

Together, these changes increased the likelihood that returned recordings would contain valid device-generated timestamps and that participant-reported timing information would be available when clock-related issues occurred.

12.3 Physical Metadata Redundancy

Stage 1 and Stage 2 processing demonstrated that handwritten notes included with returned microSD cards often provided enough supplemental information to recover or interpret datasets when online metadata were incomplete or missing.

As a result, physical handwritten metadata submission was formalized as a required component of the 2024 protocol. Participants were required to include key information such as recording location and recording start time within the return envelope alongside the microSD card.

This low-cost workflow modification created an important metadata redundancy layer that improved the recoverability and interpretability of returned datasets.

12.4 Device Preparation, Battery Handling, and Verification Workflow

The 2023 kit preparation workflow was highly manual and time-intensive, with a single team member responsible for device configuration, kit assembly, and shipment preparation. This approach became increasingly difficult to scale as participation expanded for the 2024 eclipse campaign.

For the 2024 deployment, preparation responsibilities were divided across two team members and expanded to include additional verification procedures. Project-distributed devices were also prepared with batteries pre-installed, allowing the ES team to set and verify device clocks prior to shipment.

One team member was responsible for technical preparation of each AudioMoth device. This process included:

- flashing the device to firmware version 1.8.1 using the AudioMoth Flash App
- installing batteries
- setting the device clock using the AudioMoth Time App while connected via USB
- creating a short (~10 second) test recording

Device configuration was then verified using the AudioMoth Config App. This verification process confirmed:

- the correct firmware version through review of the CONFIG.TXT file
- proper device timestamp configuration through review of the test recording timestamp

After verification, the temporary test recording was deleted.

A second team member was responsible for kit assembly and performed an additional validation step by running the ES Time Chime and visually confirming the expected LED response behavior. Devices that failed verification were removed from the deployment pool for review and either reconfigured or repurposed for non-deployment uses such as demonstrations or training.

These workflow changes improved scalability, reduced configuration errors, increased timestamp reliability, and improved consistency across deployed recording devices.

12.5 Packaging and Shipping Workflow Improvements

The 2023 shipping workflow required repeated in-person visits to USPS locations, where individual kits were manually processed and mailed. Even at the smaller 2023 deployment scale, this process proved time-intensive and operationally difficult to scale.

For the 2024 deployment, the ES team transitioned to a pre-paid shipping label workflow generated through USPS online tools. Participant mailing information stored in the Master Data

Collectors Spreadsheet was used to generate labels in advance, which were then affixed during kit assembly.

This approach reduced shipment preparation time and improved scalability for higher-volume kit distribution. USPS pickup was also coordinated for larger shipment batches, reducing the need for repeated in-person drop-offs.

Together, these workflow modifications improved the scalability, reliability, accessibility, and operational consistency of the Eclipse Soundscapes data collection system following lessons learned during the 2023 deployment.

13. Glossary

This glossary is shared across the Eclipse Soundscapes Stage 0–3 data management documents and defines terminology used throughout the ES data lifecycle workflow.

Term	Definition
AME / ES AMES	ES AudioMoth Metadata Extractor Suite. A software tool used to extract and evaluate embedded AudioMoth metadata, including timestamps, serial numbers, firmware information, and recording details to support validation and provenance workflows.
annular solar eclipse	when the Moon blocks the center of the Sun from view. The Moon appears to be too small to completely block the Sun.
API	Application Programming Interface. A structured method that allows software systems to communicate programmatically with external platforms or services.
Arbimon	An acoustic data platform originally considered by Eclipse Soundscapes for data storage, browsing, and analysis prior to the project's transition to Zenodo.
AudioMoth	An open-source, low-cost acoustic recording device used by ES volunteers to capture environmental soundscapes during eclipse events.
AZUS	Automated Zenodo Upload Software. A configuration-driven, open-source software pipeline developed by ARISA Lab to prepare, package, document, validate, and upload Eclipse Soundscapes datasets to Zenodo.
batch processing	A workflow approach that processes large groups of datasets simultaneously rather than handling one dataset at a time.
CONFIG.TXT	An AudioMoth-generated configuration file containing device settings such as sample rate, gain, firmware version, serial number, recording schedule, temperature, and battery state.
CSV	Comma-Separated Values. A simple, human- and machine-readable text file format used for spreadsheets and structured tabular data.
data dictionary	A structured reference file defining variable names, descriptions, data types, units, sources, code meanings, and notes associated with a dataset or metadata file.
dataset DOI / data site DOI	A persistent Digital Object Identifier assigned to a published Zenodo dataset record, allowing the dataset to be cited, accessed, and reused over time.

dry-run mode	A validation mode used by AZUS to verify configuration settings, metadata, credentials, and file paths before live uploads occur.
eclipse contact times	The calculated timing of eclipse phases for a specific site, including first contact, second contact, maximum eclipse, third contact, and fourth contact.
eclipse maximum	The moment of greatest solar obscuration at a specific location during a solar eclipse.
eclipse percent	The percentage of solar obscuration at a specific location.
ES ID #	Eclipse Soundscapes Identification Number. A unique numeric identifier assigned to each recording site, allowing datasets to be publicly shared without exposing participant identity.
ES WAVES	Eclipse Soundscapes WAV Audio Validation & Extraction Suite. A software system that ingests audio data from multiple microSD cards simultaneously, validates WAV files, extracts metadata, organizes datasets, and supports downstream analysis workflows.
FAIR principles	A framework for scientific data stewardship emphasizing that datasets should be Findable, Accessible, Interoperable, and Reusable.
file_list.csv	A manifest file included in each Zenodo record that lists every file, file size, file type, description, SHA-512 hash, and associated data dictionary.
first contact	solar eclipse Start (First Contact). The time at which the Moon first begins to obscure the Sun.
fourth contact	Eclipse End (Fourth Contact) The time at which the Moon completely stops blocking the Sun from view.
Hash / SHA-512 Hash	A cryptographic fingerprint used to verify file integrity. ES used SHA-512 hashes to confirm that files were not altered or corrupted during storage, transfer, or download.
Human-In-The-Loop review	A workflow step requiring manual human review and interpretation when automated validation systems identify incomplete, inconsistent, or ambiguous data conditions.
HTML README template	A reusable template used to generate standardized, site-specific Zenodo record descriptions and README documentation through automated metadata substitution.

InvenioRDM	The open-source digital repository framework underlying Zenodo. AZUS communicates with Zenodo through the InvenioRDM REST API.
JBOD	"Just a Bunch of Disks." A direct-attached storage system consisting of multiple hard drives used for scalable local data storage.
JSON configuration file	A human-readable configuration file used by AZUS to define project metadata, upload settings, templates, required metadata fields, file paths, and workflow behavior.
Level 0 data	Raw, unprocessed data preserved in its original form without timestamp correction, resampling, filtering, or modification.
metadata	Structured descriptive information about data, including location, timestamps, eclipse parameters, device information, and processing documentation.
metadata reconciliation	The process of comparing and integrating metadata from multiple sources to resolve inconsistencies and establish validated site-level records.
microSD Card	A removable flash memory storage device used within AudioMoth recorders to store WAV audio recordings and configuration files.
participatory science	Scientific research conducted with participation from members of the public contributing observations, data collection, or analysis activities.
PII	Personally Identifiable Information. Protected personal information that could uniquely identify an individual participant.
provenance	Documentation describing the origin, processing history, validation status, and stewardship history of a dataset.
README	Human-readable documentation accompanying a dataset that explains its contents, structure, collection methods, metadata, and usage guidelines.
related identifiers	Structured citation metadata linking Zenodo records to related datasets, publications, software repositories, reports, or videos.
REST API	A web-based application programming interface that allows software systems to communicate using standardized HTTP requests.
scalability	The ability of a workflow, infrastructure system, or software tool to efficiently support increasing amounts of data, users, or processing demand.
second contact	Totality Start (for sites within the path of totality). The time at which totality begins during a total solar eclipse.

site-level dataset	A complete dataset associated with a single Eclipse Soundscapes recording location identified by one ES ID #.
staging directory	A temporary preparation directory created during AZUS processing that contains the ZIP archive and all companion files ready for upload to Zenodo.
subject labels	Standardized Zenodo metadata tags used to categorize datasets by eclipse event, analysis relevance, site type, or project designation.
template substitution	An automated process in which variable placeholders within templates are replaced with site-specific metadata values during dataset generation.

14. References and Related Resources

These references and related resources are shared across the Eclipse Soundscapes (ES) Stage 0–3 data management documents and provide supporting literature, technical resources, software references, infrastructure documentation, and related materials referenced throughout the ES data lifecycle workflow.

Core Eclipse Soundscapes Resources

Eclipse Soundscapes Zenodo Community
Public archive of Eclipse Soundscapes datasets, documentation, and related resources.
https://zenodo.org/communities/eclipsesoundscapes

ARISA Lab Eclipse Soundscapes GitHub Repository
Open-source software, workflows, processing tools, and supporting documentation for the Eclipse Soundscapes project. https://github.com/ARISA-Lab-LLC/ESCSP

Eclipse Soundscapes Technical Workflow References

Severino, M., & Winter, H. (2026). Eclipse Soundscapes Data Collector Role Training and Implementation Manual (2023–2024). Zenodo. https://doi.org/10.5281/zenodo.18623443

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Winter, H. T., & Severino, M. Eclipse Soundscapes Data Management: Public Data Sharing (Stage 3). Zenodo. https://doi.org/10.5281/zenodo.18683437

Eclipse Timing and Astronomical Resources

NASA Goddard Space Flight Center Eclipse Web Site
Eclipse timing predictions and eclipse path calculations courtesy of Fred Espenak, NASA/GSFC. https://eclipse.gsfc.nasa.gov/eclipse.html

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Open Science and Repository Infrastructure Resources

Zenodo
General-purpose open-access repository operated by CERN and OpenAIRE for long-term data preservation and DOI assignment. https://zenodo.org

FAIR Principles
Findable, Accessible, Interoperable, and Reusable data stewardship principles. https://www.go-fair.org/fair-principles/

InvenioRDM
Open-source repository framework underlying Zenodo and used by AZUS through the REST API. https://inveniordm.docs.cern.ch/

Zenodo REST API Documentation
Documentation for Zenodo's Open API used by AZUS for automated dataset publication workflows.
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Appendix 1: ES Operational Infrastructure and Data Stewardship Workflows